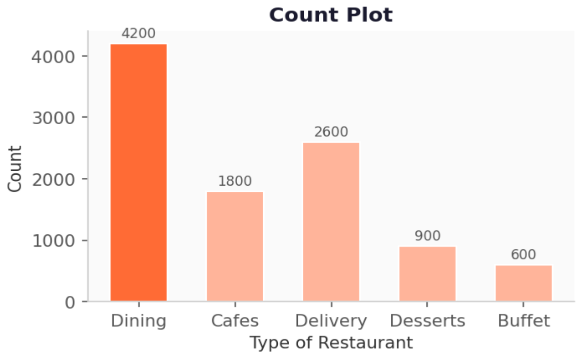
Zomato EDA Project

Exploratory Data Analysis · Teaching Notes

■ Burnout Batch

Visualization Types

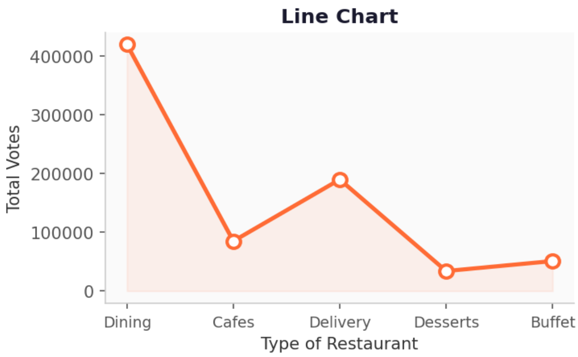


Count Plot

Counts how many times each category appears. Use when comparing groups.

```
sns.countplot(x=df['column'])
```

Used in → Q1, Q4

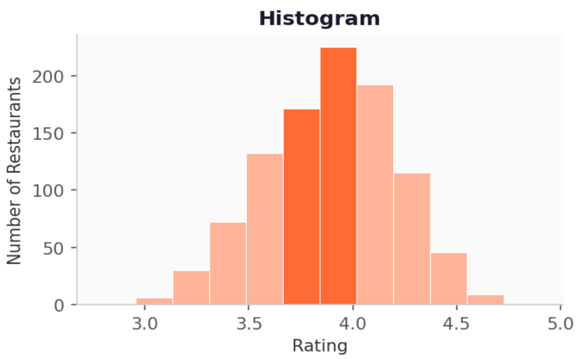


Line Chart

Connects values across categories with a line. Great for comparing totals.

```
plt.plot(data, marker='o')
```

Used in → Q2

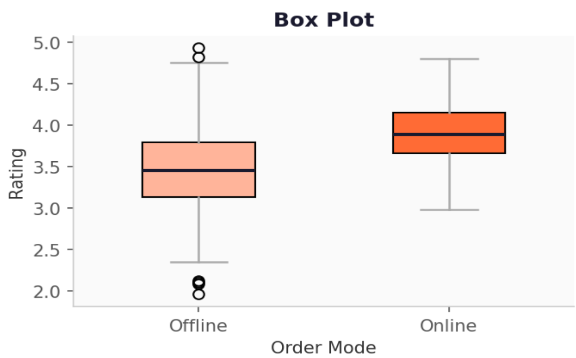


Histogram

Shows how a numeric column is distributed by grouping values into bins.

```
plt.hist(df['column'], bins=10)
```

Used in → Q3

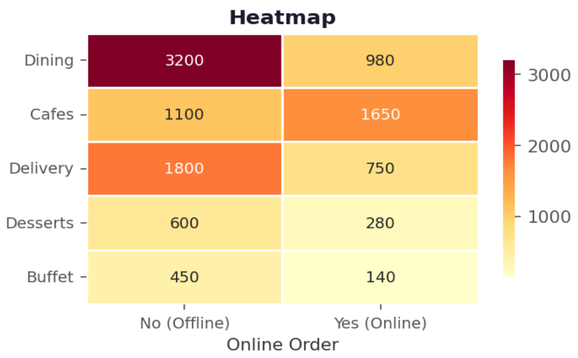


Box Plot

Shows median, spread (IQR), and outliers. Box = middle 50% of data.

```
sns.boxplot(x='cat_col', y='num_col', data=df)
```

Used in → Q5

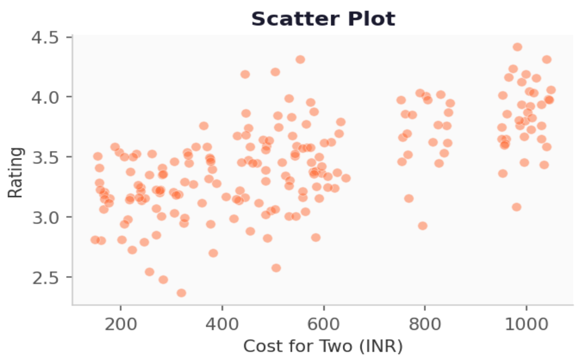


Heatmap

Colour-coded grid built from a pivot table. Darker = higher value.

```
sns.heatmap(pivot_table, annot=True, cmap='YlGnBu', fmt=
```

Used in → Q6

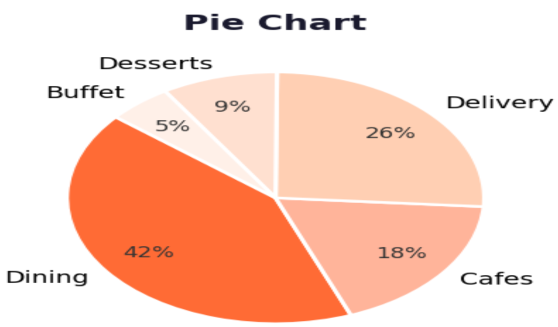


Scatter Plot

Plots two numeric columns against each other to find relationships.

```
sns.scatterplot(x='col1', y='col2', data=df)
```

Bonus — not in project



Pie Chart

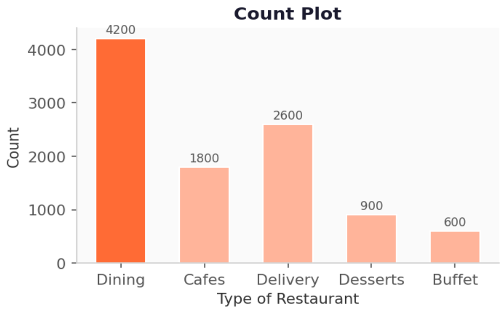
Shows proportion of each category as a slice of a whole.

```
plt.pie(values, labels=labels, autopct='%1.1f%%')
```

Bonus — not in project

The 6 Business Questions

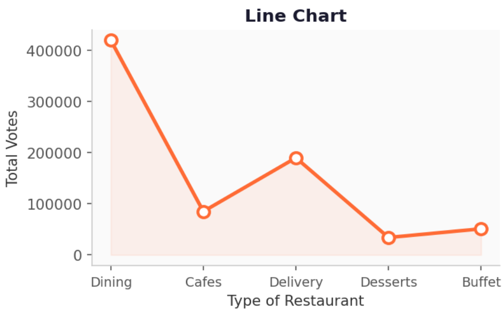
Q1 What type of restaurant do the majority of customers order from?




```
sns.countplot(x=df['listed_in(type)'])
```

✔ *Majority of restaurants fall in the Dining category.*

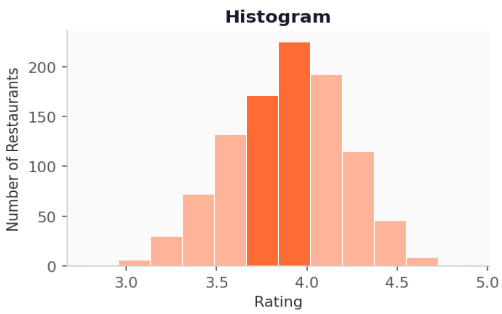
Q2 How many votes has each type of restaurant received?




```
plt.plot(result, c='green', marker='o')
```

✔ *Dining restaurants received the maximum votes.*

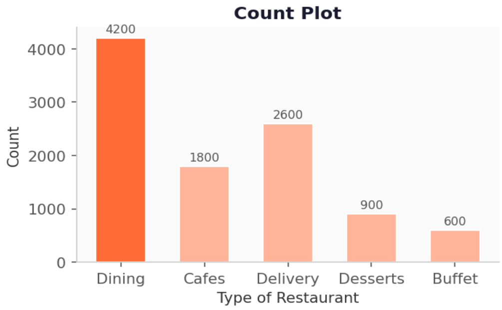
Q3 What ratings do the majority of restaurants have?




```
plt.hist(df['rate'], bins=10)
```

✔ *Most restaurants are rated between 3.5 and 4.0.*

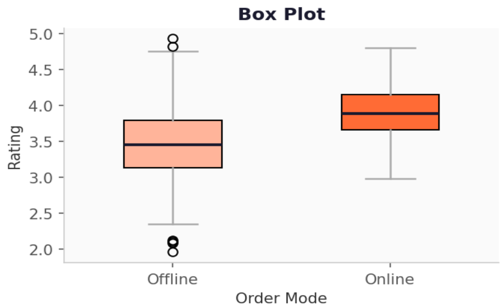
Q4 What is the average spending by couples?




```
sns.countplot(x=df['approx_cost(for two people)'])
```

✔ Most couples prefer restaurants around ■300.

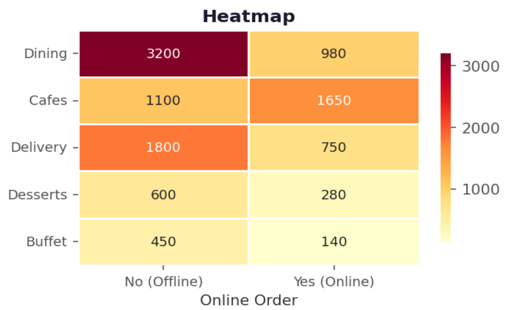
Q5 Which order mode — online or offline — gets better ratings?




```
sns.boxplot(x='online_order', y='rate', data=df)
```

✔ Online orders receive higher ratings than offline orders.

Q6 Which restaurant type gets the most offline orders?




```
sns.heatmap(pivot_table, annot=True, cmap='YlGnBu', fmt='d')
```

✔ Dining = mostly offline. Cafes = mostly online orders.

Quick Reference Sheet

Pandas

Command	What it does
`df.head(5)`	Preview first 5 rows
`df.info()`	Column names, types, null counts
`df['col'].apply(fn)`	Apply a function to every value in a column
`df.groupby('col')['v'].sum()`	Sum values grouped by a category
`df.pivot_table(index, columns, aggfunc)`	Build a cross-tab matrix for heatmap
`df['col'].isnull().sum()`	Count missing values

Seaborn

Function	Chart	Key Params
`sns.countplot()`	Count Plot	x=
`sns.boxplot()`	Box Plot	x=, y=
`sns.heatmap()`	Heatmap	annot=True, cmap=, fmt='d'
`sns.scatterplot()`	Scatter	x=, y=
`sns.histplot()`	Histogram	x=, bins=, kde=
`sns.barplot()`	Bar Plot	x=, y=, estimator=

Matplotlib

Command	Purpose
`plt.xlabel('text')`	Label the x-axis
`plt.ylabel('text')`	Label the y-axis
`plt.title('text')`	Add a chart title
`plt.figure(figsize=(w,h))`	Set chart size in inches
`plt.show()`	Render and display
`plt.hist(col, bins=n)`	Histogram with n bars
`plt.plot(data, marker='o')`	Line chart with dot markers